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Impedance monitoring of a modular electrolysis system

Abstract

An alternating current (AC) impedance spectroscopy method includes providing an AC impedance spectroscopy ripple from power electronics into an electrochemical device, and absorbing the ripple in the power electronics.

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Background/Summary

FIELD

(1) The present disclosure is directed to health and performance monitoring and control for an electrochemical system, such as a proton exchange membrane (PEM) stack system using alternating current impedance spectroscopy derived signals.

BACKGROUND

(2) Electrical circuits in which a cell is either a power source or a load are problematic in that chemical degradation effects can take place for which the only observable symptom or measurement is a degradation in the potential or impedance of the cell. This is problematic for systems implemented in field where the degrading condition is occurring and the only measurement is voltage because there is no way to differentiate between possible failure modes that can cause the degradation and to remedy the degrading condition before the degrading condition becomes permanent or chronic. Alternating current impedance spectroscopy can be used in a laboratory setting to analyze different frequency responses of cells to determine different failure modes. However, for applications in the field, laboratory equipment may not be used because it is not hardened and the resultant ripple from a laboratory signal generator if it were used at a kilowatt or even megawatt scale would be so impactful to use case customers that it would not be permitted.

SUMMARY

(3) According to one embodiment, an alternating current (AC) impedance spectroscopy method includes providing an AC impedance spectroscopy ripple from power electronics into an electrochemical device, and absorbing the ripple in the power electronics.

(4) According to another embodiment, an alternating current (AC) impedance spectroscopy system includes a power factor corrected (PFC) rectifier electrically connectable to an AC power source via an AC split bus and to a direct current DC/DC converter via a DC bus. The PFC rectifier includes a plurality of electric power inverters electrically connectable to respective AC busses of the AC split bus, a first capacitor and a second capacitor electrically connected in parallel to the plurality of electric power inverters via a middle DC bus at a DC-link midpoint; and an auxiliary electric power converter electrically connected to the DC-link midpoint via the middle DC bus and electrically connectable to the direct current (DC)/DC converter via the DC bus.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIGS. **1**A and **1**B are circuit block diagrams illustrating power electronics hardware configured for alternating current (AC) impedance spectroscopy including a power factor corrected (PFC) rectifier configured for ripple cancelation according to a first embodiment.

(2) FIGS. **2**A and **2**B are circuit block and control flow diagrams illustrating power electronics hardware and control configured for AC impedance spectroscopy including digital control of a PFC rectifier for ripple cancelation according to the first embodiment.

(3) FIGS. **3**A and **3**B are circuit block diagrams illustrating power electronics hardware configured for AC impedance spectroscopy including a power inverter configured for ripple cancelation according to a second embodiment.

(4) FIGS. **4**A and **4**B are circuit block and control flow diagrams illustrating power electronics hardware and control configured for AC impedance spectroscopy including digital control of a power inverter for ripple cancelation according to the second embodiment.

(5) FIGS. **5**A and **5**B are circuit block diagrams illustrating power electronics hardware configured for AC impedance spectroscopy including a direct current (DC)/DC converter configured for ripple

cancelation according to alternative embodiments.
(6) FIG. **6** is a process flow diagrams illustrating an example method of impedance monitoring of a modular electrolysis system according to an embodiment.
(7) FIG. **7** is a component block diagram illustrating an example computing device suitable for implementing various embodiments.
(8) Like reference symbols in the various drawings indicate like elements.
DETAILED DESCRIPTION
(9) The embodiments will now be described more fully hereinafter with reference to the accompanying figures, in which exemplary embodiments are shown. The foregoing may, however, be embodied in many different forms and should not be construed as limited to the exemplary embodiments set forth herein. All fluid flows may flow through conduits (e.g., pipes and/or manifolds) unless specified otherwise.
(10) All documents mentioned herein are hereby incorporated by reference in their entirety. References to items in the singular should be understood to include items in the plural, and vice versa, unless explicitly stated otherwise or clear from the text. Grammatical conjunctions are intended to express any and all disjunctive and conjunctive combinations of conjoined clauses, sentences, words, and the like, unless otherwise stated or clear from the context. Thus, the term “or” should generally be understood to mean “and/or,” and the term “and” should generally be understood to mean “and/or.”
(11) Recitation of ranges of values herein are not intended to be limiting, referring instead individually to any and all values falling within the range, unless otherwise indicated herein, and each separate value within such a range is incorporated into the specification as if it were individually recited herein. The words “about,” “approximately,” or the like, when accompanying a numerical value, are to be construed as including any deviation as would be appreciated by one of ordinary skill in the art to operate satisfactorily for an intended purpose. Ranges of values and/or numeric values are provided herein as examples only, and do not constitute a limitation on the scope of the described embodiments. The use of any and all examples or exemplary language (“e.g.,” “such as,” or the like) is intended merely to better illuminate the embodiments and does not pose a limitation on the scope of those embodiments. No language in the specification should be construed as indicating any unclaimed element as essential to the practice of the disclosed embodiments.
(12) The embodiments described herein reference a “cell” as an electric power source and/or an electric load of an electrochemical system, such as a proton exchange membrane (PEM) fuel cell generator, PEM electrolyzer, other PEM unit, and/or other electrochemical system, which may include cells, such as electrochemical cells (e.g., batteries, flow batteries, fuel cells, electrolyzer cells, hydrogen pumping cells, oxygen pumping cells), photovoltaic cells, motors, generators, and other semiconductor junction based cells. In some embodiments, a cell as an electric power source may include electrical or electrochemical ammonia synthesis cells, electrical or electrochemical metal-organic framework cell (MOF-cell) water condenser, electrical or electrochemical oxygen, nitrogen, or air purifiers, electrical or electrochemical water polisher, such as an electrodialysis cell, etc. In some embodiments, a cell as an electric load may include a customer load, (e.g., an electrical system of a building, a facility, an environment conditioning system, a machine, a computer network, etc.), a hydrogen pumping cell, and oxygen generating cell, a MOF-cell, etc. Examples of such PEM electrolyzer cells are described in U.S. patent application Ser. No. 17/101,224, entitled “SYSTEMS AND METHODS OF AMMONIA SYNTHESIS” by Ballantine et al., U.S. patent application Ser. No. 17/101,232, entitled “ELECTROCHEMICAL DEVICES, MODULES, AND SYSTEMS FOR HYDROGEN GENERATION AND METHODS OF OPERATING THEREOF” by Ballantine et al., and U.S. Provisional Patent Application No. 63/057,406, entitled “MODULAR SYSTEM FOR HYDROGEN AND AMMONIA GENERATION WITHOUT DIRECT WATER INPUT FROM CENTRAL SOURCE” by Light et

al. the entire contents of each of these references incorporated herein by reference.
(13) Embodiments and examples herein may be described with reference to PEM based electrolyzer stacks (herein PEM stack) for clarity and ease of explanation, and do not limit the scope of the claims and descriptions to PEM stacks as one of skill in the art shall recognize that such embodiments may be similarly applicable to other cells.
(14) Embodiments provide devices and methods of determining the health and points of control for an electrochemical system, such as a PEM stack based system, using alternating current (AC) impedance spectroscopy derived signals. Such devices and methods may be configured to monitor for leakage, errors, deterioration, and overall life quality of a PEM stack using a small AC ripple across a direct current (DC) power input into the PEM stack and reading the frequency of the resultant impedance wave. In some embodiments the AC ripple may sweep through set frequencies and gather data to analyze against predetermined errors and the system may be corrected to clear the error and/or the error may be reported to the owner/customer. The small ripple current should not be higher than the DC current magnitude and should not impact the PEM stack operation or life.
(15) Existing circuit arrangements for AC impedance spectroscopy require the power flow to or from the electrochemical stack to be interrupted and impedance to be monitored by observing the voltage transient response. This method has the substantial disadvantage of requiring an interruption of the power flow. Further, the full breadth of frequency spectrum analysis is not possible with this method.
(16) Existing circuit arrangements for AC impedance spectroscopy also include structures where multiple elements of power electronics are operated 180 degrees out of phase to cancel ripple on the power bus when ripple is applied to the electrochemical stack bus. This has the substantial disadvantage of requiring that multiple circuits be operated in order to create the canceling effect. A single stack circuit is not possible.
(17) Embodiments herein provide power electronics hardware and power electronics control configured for in-situ AC impedance spectroscopy in which the ripple back to a power supply source may be cancelled. Such embodiments avoid the requirement of existing circuit arrangements for AC impedance spectroscopy for multiple segments of power electronics to be operating out of phase to cancel ripple currents.
(18) Embodiments herein provide integration of in-situ AC impedance spectroscopy in an electrochemical system, such as a PEM fuel cell generator, PEM electrolyzer, other PEM unit or other electrochemical system, where the feature of AC impedance spectroscopy is used for system diagnostics and control.
(19) FIGS. **1**A-**5**B illustrate various embodiments of power supply electronics, located in a power supply module of an electrochemical system, such as a PEM fuel cell generator, PEM electrolyzer, other PEM unit or other electrochemical system. The power supply power electronics may be configured to create a specified AC current ripple, superimposed on a DC current supplied to a cell, for, example an electrolyzer stack, such as a PEM stack, in an electrolyzer module.
(20) The resulting ripple back to a power supply source may be canceled so that there is no resulting impact of harmonics to a load for the electrochemical system. In some embodiments, the ripple may be cancelled by ripple canceling power electronics hardware, as described further herein with reference to FIGS. **1**A, **1**B, **3**A, and **3**B. In some embodiments, the ripple may be cancelled by ripple canceling controls inside an error correcting loop of the power electronics, as described further herein with reference to FIGS. **2**A, **2**B, **4**A, and **4**B. In some embodiments, the ripple may be cancelled by ripple canceling switches, as described further herein with reference to FIGS. **5**A and **5**B.
(21) A voltage on the electrolyzer stack or groups of cells may monitored by electronics in the power supply module and/or in the electrolyzer module. A phasor impedance of the electrolyzer stack at the ripple frequency may be determined by comparing a driving current ripple signal with

an induced voltage ripple. Multiple impedances at different frequencies may be determined by changing the driving current ripple frequency. In some embodiments, the driving current ripple may be swept through frequencies in a stepwise fashion. In some embodiments, the driving current ripple may be selected frequencies used individually or in small sets to obtain specific impedance data at specific frequencies.

(22) Based on the impedance data, controls may be executed in order to improve the performance and lifetime of the electrolyzer module. The power electronics hardware and power electronics controls described herein may be integrated in tandem with system level controls for diagnostics or function control of the electrochemical system.

(23) FIGS. **1**A and **1**B are circuit block diagrams illustrating power electronics hardware **100***a*, **100***b* configured for AC impedance spectroscopy including a power factor corrected (PFC) rectifier **102** configured for ripple cancelation according to a first embodiment. With reference to FIGS. **1**A and **1**B, the power electronics hardware **100***a*, **100***b* may be configured to provide electric power to an electrochemical system, such as a PEM fuel cell generator, PEM electrolyzer, other PEM unit or other electrochemical system. The power electronics hardware **100***a*, **100***b* may include the PFC rectifier **102** and a DC/DC converter **104**, which may be electrically coupled to other components (not shown) of the power electronics hardware **100***a*, **100***b*, such as a cell as an electric power source, and/or a cell as an electric load (not shown). The power electronics hardware **100***a*, **100***b* may further include or be electrically connectable to an AC power source **120**, which may provide a 3-phase AC via an AC bus **121**. The AC power source **120** may include a generator, an electric grid, a load bank, etc. The AC power source **120** may be electrically connectable to the PFC rectifier **102** via multiple inductors **122**. For example, an inductor **122** may be electrically connectable between different phase outputs of the AC power source **120** and the PFC rectifier **102**, such as at least one inductor **122** electrically connectable to a bus of a split AC bus **121**. The AC power source **120** may also be electrically connectable to a ground reference **132**.

(24) The PFC rectifier **102** may include various components configured for electric power conditioning and for AC impedance spectroscopy. The various components may include any number of electric power inverters **106**, which may be electrically connectable to the AC power source **120**, such as each of the electric power inverters **106** may be electrically connectable to the AC bus of the split AC bus **121** and at least one respective inductor **122**. The PFC rectifier **102** may be electrically connected to the DC/DC converter **104**. For example, the electric power inverters **106** may include at least two diodes, such as a first diode **118***a* and a second diode **118***b*, and the respective bus of the AC split bus **121** may be electrically connectable between an anode end of the first diode **118***a* and a cathode end of the second diode **118***b*. The cathode end of the first diode **118***a* may be electrically connected to a DC bus **123***a* and the anode end of the second diode **118***b* may be electrically connected to a return DC bus **123***b*. In some embodiments, the DC bus **123***a* and the return DC bus **123***b* may have opposite polarities.

(25) In some embodiments, a half-bridge inverter **112** may be incorporated into each of the electric power inverters **106**. A first end of the half-bridge inverter **112** may be electrically connected between the anode end of the first diode **118***a* and the cathode end of the second diode **118***b*. As such, in some embodiments, the first end of the half-bridge inverter **112** may be electrically connectable to the respective bus of the AC split bus **121** connected between the anode end of the first diode **118***a* and the cathode end of the second diode **118***b*. The half-bridge inverter **112** may be electrically connected to a middle DC bus **123***c*. In some embodiments, the middle DC bus **123***c* may be a neutral bus. In some embodiments, the half-bridge inverter **112** may include opposingly oriented pairs of paralleled diodes **114***a*, **114***b* and transistors **116***a*, **116***b*, such as insulated-gate bipolar transistors (IGBTs) or metal-oxide-semiconductor field-effect transistors (MOSFETs). The pairs of paralleled diodes **114***a*, **114***b* and transistors **116***a*, **116***b* may include a cathode end of a diode **114***a*, **114***b* electrically connected to a collector end of a transistor **116***a*, **116***b*, and an anode end of the diode **114***a*, **114***b* electrically connected to an emitter end of the transistor **116***a*, **116***b*.

The cathode end of a first diode **114***a* and the collector end of a first transistor **116***a* may be electrically connected in parallel between the anode end of the first diode **118***a* and the cathode end of the second diode **118***b*. The anode ends of the diodes **114***a*, **114***b* and the emitter ends of the transistors **116***a*, **116***b* may be electrically connected in parallel to a bus (e.g., bus **122**). The transistors **116***a*, **116***b* may each include a gate end electrically connected in parallel to a bus. The cathode end of a second diode **114***b* and the collector end of a second transistor **116***b* may be electrically connected in parallel to the middle DC bus **123***c*.
(26) The PFC rectifier **102** may include at least two capacitors, such as a first capacitor **108***a* and a second capacitor **108***b*, electrically connected in parallel to the middle DC bus **123***c*. The first capacitor **108***a* may be electrically connected to the middle DC bus **123***c* at a cathode end and electrically connected to the DC bus **123***a* at an anode end. The second capacitor **108***b* may be electrically connected to the middle DC bus **123***c* at an anode end and electrically connected to the return DC bus **123***b* at a cathode end. The capacitors **108***a*, **108***b* may be electrically connected in parallel to the electric power inverters **106** and/or the half-bridge inverters **112** via the middle DC bus **123***c*. In some embodiments, the capacitors **108***a*, **108***b* may be configured to approximately equally divide the DC voltage between the electric power inverters **106** and an auxiliary electric power converter **110** of the PFC rectifier **102**. A connection of the capacitors **108***a*, **108***b* to the middle DC bus **123***c* may be a midpoint (“0”) for a DC-link of the PFC rectifier **102**.
(27) The auxiliary electric power converter **110** may include at least two additional capacitors, such as a first capacitor **124***a* and a second capacitor **124***b*, electrically connected in parallel to the middle DC bus **123***c*. The first capacitor **124***a* may be electrically connected to the DC bus **123***a* at a first end and electrically connected to the middle DC bus **123***c* at a second end. The second capacitor **124***b* may be electrically connected to the middle DC bus **123***c* at a first end and electrically connected to the return DC bus **123***b* at a second end. The capacitors **124***a*, **124***b* may be electrically connected in parallel to a filter inductor **126**. The filter inductor **126** may be electrically connected to the middle DC bus **123***c* at a first end and to a second half-bridge inverter at a second end. The second half-bridge inverter has pairs of paralleled transistors **128***a*, **128***b*, such as insulated-gate bipolar transistors (IGBTs) or metal-oxide-semiconductor field-effect transistors (MOSFETs), and diodes **130***a*, **130***b*. The pairs of paralleled transistors **128***a*, **128***b* and diodes **130***a*, **130***b* may include a collector end of a transistor **128***a*, **128***b* electrically connected to a cathode end of a diode **130***a*, **130***b*, and an emitter end of the transistor **128***a*, **128***b* electrically connected to an anode end of the diode **130***a*, **130***b*. The collector end of a first transistor **128***a* and the cathode end of a first diode **130***a* may be electrically connected in parallel to the DC bus **123***a*. The emitter end of the first transistor **128***a* and the anode end of the first diode **130***a* may be electrically connected in parallel to the second end of the filter inductor **126**. The collector end of a second transistor **128***b* and the cathode end of a second diode **130***b* may be electrically connected in parallel to the second end of the filter inductor **126**. The emitter end of the second transistor **128***b* and the anode end of a second diode **130***b* may be electrically connected in parallel to the return DC bus **123***b*.
(28) A DC/DC converter **104** may be electrically connected between the DC bus **123***a* and the return DC bus **123***b*. The DC/DC converter **104** may be electrically connectable to any number and combination of other components (not shown) of the power electronics hardware **100***a*, **100***b*, such as a cell as an electric power source, and/or a cell as an electric load (not shown).
(29) In some embodiments, a DC-link midpoint (“O”) of the PFC rectifier **102** may exist at a connection of the capacitors **108***a*, **108***b* and the middle DC bus **123***c*. Depending on the state of charge of the capacitors **108***a*, **108***b*, the DC-link midpoint's availability may vary. When the DC-link midpoint is not accessible, as shown in FIG. **1**B, the capacitors **124***a*, **124***b* be may be actively connected to the DC middle bus **123***c* instead of to the DC bus **123***a* and the DC return bus **123***b* for use in AC impedance spectroscopy.
(30) The auxiliary electric power converter **110** may be switched, for example, through control of

the transistors **130**_a_, **130**_b_ in such a way that the auxiliary electric power converter **110** may draw only reactive power from the DC-link, with a specified magnitude and phase angle. The current drawn by the auxiliary electric power converter **110** from the DC-link may be controlled to compensate for a ripple current drawn by the DC/DC converter **104**, from the DC-link of the PFC rectifier **102**. In some embodiments control of the transistors **128**_a_, **128**_b_ and the diodes **130**_a_, **130**_b_ may be implemented using any number and combination of controllers (not shown).
(31) FIGS. **2**A and **2**B are circuit block and control flow diagrams illustrating power electronics hardware **200** and control **210** configured for AC impedance spectroscopy including digital control of a PFC rectifier **202** for ripple cancelation according to the second embodiment. With reference to FIGS. **1**A-**2**B, like reference numbers in FIG. **2**A and FIGS. **1**A and **1**B may be similarly described, including: the ground reference **132**; the AC power source **120**; the inductors **122**; the electric power inverters **106**; the diodes **118**_a_, **118**_b_; the half-bridge inverter **112**; the capacitors **108**_a_, **108**_b_; and the DC/DC converter **104**. The reference numbers for components of the half-bridge inverter **112** have been omitted for the sake of simplicity and clarity of the drawings. However, components of the half-bridge inverter **112**, such as the diodes **114**_a_, **114**_b_ and the transistors **116**_a_, **116**_b_, as illustrated in and described with reference to FIGS. **1**A and **1**B, may be similarly described with reference to FIGS. **2**A and **2**B.
(32) In some embodiments, the control of the power electronics hardware **200** configured for AC impedance spectroscopy may be implemented by any number and combination of controllers (not shown). For example, a controller may receive signals from the AC power source **120** ("A"), signals from the inductors **122** ("B"), and signals from the capacitors **108**_a_, **108**_b_ ("C" and "D"). The signals (A-D) may be representative of voltage and/or current levels at each of the AC power source **120**, the inductors **122**, and the capacitors **108**_a_, **108**_b_. The controller may use the signals (A-D) to implement control of the power electronics hardware **200** configured for AC impedance spectroscopy, as illustrated, for example, in FIG. **2**B.
(33) In the PFC rectifier **202**, current loops may operate at high (e.g., several kHz) bandwidth to track current reference commands effectively. Typically, a DC bus voltage control loop would have a lower bandwidth to maintain sinusoidal grid currents, for example at the AC power source **120** ("A") and/or at the inductors **122** ("B"), by providing appropriate current reference command (e.g., "S.sub.A", "S.sub.B", "S.sub.C"). In other words, the DC bus voltage loop bandwidth will be lower when compared to the current loop. The higher and lower bandwidths are relative and depend on the power rating of the underlying equipment.
(34) A ripple voltage on the DC-link of the PFC rectifier **202**, due to the converter perturbation for AC impedance spectroscopy at the DC/DC converter **104**, may appear at an input error signal of a voltage controller. When the voltage controller error input contains low frequency signals (e.g., signals whose frequency is close to the voltage loop bandwidth or higher, especially around the voltage loop bandwidth), a ripple portion in the error signal may reach an output of the voltage controller. This adds undesirable frequency components to the current reference command. A current controller may ensure proper tracking of the reference command, which may result in injection of a lower frequency signal (related to the frequency perturbation injected by the DC/DC converter **104**) into an electric power grid, such as AC power source **120**.
(35) To mitigate injection of the low frequency components to the grid, a controller may estimate or find a DC-link ripple voltage based on the known electrical quantities and injected perturbation for the AC impedance spectroscopy, or the controller may track the ripple voltage on the DC-link. The controller may remove the ripple component from a sensed DC-link voltage signal or error input to the controller, such as a DC-link voltage controller. This may prevent propagation of a perturbed voltage ripple in the DC-link from reaching the controller, such as the current controller, and effectively prevent the injection of the low frequency components into the grid.
(36) The controller may implement a number of operations to remove the ripple component from a sensed DC-link voltage signal or error. For example, the controller may receive signals from the

AC power source **120** ("A"), signals from the inductors **122** ("B"), and signals from the capacitors **108***a*, **108***b* ("C" and "D"). These signals (A-D) may be voltage signals at the respective components. The controller may sum **212***a*, **212***h* the signals (C and D), for example to generate a total voltage and a voltage difference of the signals (C and D). The controller may also receive three phase signals (A and B) and convert **218***a*, **218***b* three phase signals to voltage and current components in a two-axis reference frame (e.g., d-q). A hardware zero-crossing or digital phase-locked loop operation **224** may determine an angular frequency for converting **218***a*, **218***b* three phase signals to voltage and current components. A voltage sensor-less control also can be used for determining the grid phase angle or angular frequency.

(37) The total voltage may be further summed **212***b* with a reactive voltage component to generate a real voltage component. The real voltage component may be summed **212***c* with a reference real voltage component to generate a voltage difference of the real voltage, which may be converted **214***a* (e.g., by a proportional and integral ("PI") operation) to generate a corresponding reference current.

(38) Reference currents in the two-axis reference frame may be summed **212***d*, **212***f* with the current components in the two-axis reference frame converted from the three phase signal (B) to generate a current differences, of which differentials **214***b*, **214***c* (e.g., by a PI operation) may be generated. In some embodiments, a reactive power reference current, summed **212***f* with a reactive power current, may equal zero. The differentials may be summed **212***e*, **212***g* with portions of the voltage components in the two-axis reference frame converted from the three phase signal (A) to generate values in the two-axis reference frame. The voltage difference of the signals (C and D) may be summed **212***i* with a zero value, from the result of which a value may be generated **214***d* (e.g., by a PI operation) in a third axis reference. The resultant values may be converted **216** to values in a three-axis reference frame of the received signals (A and B). The values in the three-axis reference frame may be augmented **220***a*, **220***b*, **220***c* by a reference voltage and converted **222** (e.g., by an enhanced pulse width modulation ("ePWM") operation) to generate current reference commands (e.g., "S.sub.A", "S.sub.B", "S.sub.C") that may be provided to the PFC rectifier **202**.

(39) In some embodiments in which an AC power source is a cell such as an electrochemical cell, or generator, and power electronics pass this power to a load, and in which the power electronics impart a perturbation, ripple, or sinusoid in sourced current on the cell, device, electrochemical cell or generator such that by monitoring the voltage of the cell or generator, the impedance of that item at the frequency of the perturbation, ripple, or sinusoid may be determined. Further, the power electronics may be configured to cause direct cancelation of the perturbation, ripple, or sinusoid such that it does not pass downstream into a feed to power to a load.

(40) FIGS. **3**A and **3**B are circuit block diagrams illustrating power electronics hardware **300***a*, **300***b* configured for AC impedance spectroscopy including a power inverter configured for ripple cancelation according to a second embodiment. With reference to FIGS. **1**A-**3**B, like reference numbers in FIGS. **3**A and **3**B, and FIGS. **1**A-**2**B may be similarly described, including: the ground reference **132**; the AC power source **120**; the inductors **122**; the electric power inverters **106**; the capacitors **108***a*, **108***b*; the auxiliary electric power converter **110**; the capacitors **124***a*, **124***b*; the filter inductor **126**; the transistors **128***a*, **128***b*; the diodes **130***a*, **130***b*; and the DC/DC converter **104**.

(41) The electric power inverters **106** may include at least two pairs of paralleled transistors **308***a*, **308***b*, such as insulated-gate bipolar transistors (IGBTs) or metal-oxide-semiconductor field-effect transistors (MOSFETs), and diodes **310***a*, **310***b*, instead of the diodes **118***a* and **118***b* of the first embodiment shown in FIGS. **1**A and **1**B. The pairs of paralleled transistors **308***a*, **308***b* and diodes **310***a*, **310***b* may include a collector end of a transistor **308***a*, **308***b* electrically connected to a cathode end of a respective diode **310***a*, **310***b*, and an emitter end of the transistor **308***a*, **308***b* electrically connected to an anode end of the respective diode **310***a*, **310***b*. The collector end of a first transistor **308***a* and the cathode end of a first diode **310***a* may be electrically connected in

parallel to the DC bus **123***a*. The emitter end of the first transistor **308***a* and the anode end of the first diode **310***a* may be electrically connected in parallel to a respective bus of the AC split bus **121**. The collector end of a second transistor **308***b* and the cathode end of a second diode **310***b* may be electrically connected in parallel to the respective bus of the AC split bus **121**. The emitter end of the second transistor **308***b* and the anode end of a second diode **310***b* may be electrically connected in parallel to the return DC bus **123***b*.

(42) In some embodiments, a half-bridge inverter **302** may be incorporated into each of the electric power inverters **106**. A first end of the half-bridge inverter **302** may be electrically connected between the emitter end of the first transistor **308***a* and the anode end of the first diode **310***a*, and the collector end of the second transistor **308***b* and the cathode end of the second diode **310***b*. As such, in some embodiments, the first end of the half-bridge inverter **302** may be electrically connectable to the respective bus of the AC split bus **122** which is electrically connected between the emitter end of the first transistor **308***a* and the anode end of the first diode **310***a*, and the collector end of the second transistor **308***b* and the cathode end of the second diode **310***b*. The second end of half-bridge inverter **302** may be electrically connected to the middle DC bus **123***c*. In some embodiments, the middle DC bus **123***c* may be a neutral bus. In some embodiments, the half-bridge inverter **302** may include pairs of paralleled transistors **306***a*, **306***b*, such as insulated-gate bipolar transistors (IGBTs) or metal-oxide-semiconductor field-effect transistors (MOSFETs), and diodes **304***a*, **304***b*. The pairs of paralleled transistors **306***a*, **306***b* and diodes **304***a*, **304***b* may include a collector end of a transistor **306***a*, **306***b* electrically connected to a cathode end of a diode **304***a*, **304***b*, and an emitter end of the transistor **306***a*, **306***b* electrically connected to an anode end of the diode **304***a*, **304***b*. The collector end of a first transistor **306***a* and the cathode end of a first diode **304***a* may be electrically connected in parallel between the emitter end of the first transistor **308***a* and the anode end of the first diode **310***a*, and the collector end of the second transistor **308***b* and the cathode end of the second diode **310***b*. The emitter end of the first transistor **306***a* and the anode end of the first diode **304***a* may be electrically connected in parallel to a bus. The collector end of a second transistor **306***b* and the cathode end of a second diode **304***b* may be electrically connected in parallel to the bus. The emitter end of the second transistor **306***b* and the anode end of a second diode **304***b* may be electrically connected in parallel to the middle DC bus. The transistors **306***a*, **306***b* may each include a gate end electrically connected in parallel to a bus.

(43) The capacitors **108***a*, **108***b* of the power electronics hardware **300***a*, **300***b* may be electrically connected in parallel to the middle DC bus **123***c*. The first capacitor **108***a* may be electrically connected to the middle DC bus **123***c* at the cathode end and the second capacitor **108***b* may be electrically connected to the middle DC bus **123***c* at the anode end. The capacitors **108***a*, **108***b* may be electrically connected in parallel to the electric power inverters **106** and/or the half-bridge inverters **302** via the middle DC bus **123***c*.

(44) The active ripple mitigation in the power electronics hardware **300***a*, **300***b* for the electric power inverters **106** may be the same as that presented for the PFC rectifier **102** in FIGS. **1**A and **1**B. When the DC-link mid-point ("O") is not accessible, the capacitors **124***a*, **124***b* may create a virtual DC-link midpoint and a wide voltage variation, close to the dc bus voltage, across the capacitors **124***a*, **124***b* may be acceptable. By controlling a ripple current magnitude and phase angle, drawn by the auxiliary electric power converter **110** from the DC bus **123***a*, a voltage ripple across the capacitors **108***a*, **108***b* may be mitigated for all inverter loading conditions at different power factors. In some embodiments control of the transistors **308***a*, **308***b*, **130***a*, **130***b* may be implemented using any number and combination of controllers (not shown).

(45) FIGS. **4**A and **4**B are circuit block and control flow diagrams illustrating power electronics hardware **400** and control **410** configured for AC impedance spectroscopy including digital control of a power inverter for ripple cancelation according to the second embodiment. With reference to FIGS. **1**A-**4**B, like reference numbers in FIG. **4**A and FIGS. **1**A-**2**A, **3**A, and **3**B may be similarly described, including: the ground reference **132**; the AC power source **120**; the inductors **122**; the

electric power inverters **106**; the transistors **308***a*, **308***b*; the diodes **310***a*, **310***b*; the half-bridge inverter **302**; the capacitors **108***a*, **108***b*; and the DC/DC converter **104**. The reference numbers for components of the half-bridge inverter **302** have been omitted for the sake of simplicity and clarity of the drawings. However, components of the half-bridge inverter **302**, such as the diodes **304***a*, **304***b* and the transistors **306***a*, **306***b*, as illustrated in and described with reference to FIGS. **3**A and **3**B, may be similarly described with reference to FIGS. **4**A and **4**B.

(46) The digital control **210** of the ripple mitigation presented for the PFC rectifier **102** in FIGS. **2**A and **2**B may be extended for the digital control **410** of the electric power inverters **106** in FIG. **4**A. The inverter **106** may be a two level or multilevel inverter. The control architecture for digital control **410** may be the same the control architecture for digital control **210**, except there may be two current references (one for real power and one for reactive power). The ripple in the DC-link may propagate to a current reference pertaining to a real power in both inverter **106** in FIG. **4**A and PFC rectifier **102** in FIG. **2**A. The reactive power in the PFC rectifier **102**, such as summed **212***f* with a real power, may always be zero in FIG. **4**B. The real and reactive power may be independent and either positive or negative for the inverter **106**, where bidirectional power flow may be possible.

(47) FIGS. **5**A and **5**B are circuit block diagrams illustrating power electronics hardware **500** including a DC/DC converter **508** and compensating set of switches configured for AC impedance spectroscopy with ripple cancelation according to alternative embodiments. With reference to FIGS. **1**A-**5**B, the power electronics hardware **500** may include a compensating set of switches, an example of which is illustrated in FIG. **5**B configured to offset a ripple created and injected into cells as an electric power source and/or cells an electric power load by AC impedance spectroscopy.

(48) The power electronics hardware **500** may include a PFC rectifier **504** (e.g., PFC rectifier **102** in FIGS. **1**A-**2**A and **3**A-**4**A) that may be electrically connectable to an AC power source **502** (e.g., AC power source **120** in FIGS. **1**A-**2**A and **3**A-**4**A). The power electronics hardware **500** may further include a first DC bus **516** electrically connecting the PFC rectifier **504** to at least two DC/DC converters **506**, **508** in parallel. In some embodiments, the first DC bus **516** maybe approximately an 800 V bus. In some embodiments, the DC/DC converter **506** (e.g., DC/DC converter **104** in FIGS. **1**A-**2**A and **3**A-**4**A) may be electrically connectable to a cell as a load **514**. The DC/DC converters **506**, **508** may be electrically connected to an AC sweep generator **510**, configured to generate any number and combination of AC signals, such as single AC signals, step AC signals, continuous ranges of AC signals, which may vary in voltage. The AC signals generated by the AC sweep generator **510** may be used by the DC/DC converter **506** to measure impedance of the cell as a load **514**. The DC/DC converter **508** may be electrically connected to the split DC bus **516** at an anode end and to a capacitor **512** at a cathode end via a second DC bus **518**. In some embodiments, the second DC bus **518** maybe approximately a 400 V bus.

(49) In some embodiments, described with reference to FIG. **5**B, the DC/DC converter **508** may include at least two pairs of paralleled transistors **520***a*, **520***b*, such as insulated-gate bipolar transistors (IGBTs) or metal-oxide-semiconductor field-effect transistors (MOSFETs), and diodes **522***a*, **522***b*. The pairs of paralleled transistors **520***a*, **520***b* and diodes **522***a*, **522***b* may include a collector end of a transistor **522***a*, **522***b* electrically connected to a cathode end of a diode **522***a*, **522***b*, and an emitter end of the transistor **520***a*, **520***b* electrically connected to an anode end of the diode **522***a*, **522***b*. The emitter end of a first transistor **520***a* and the anode end of a first diode **522***a* may be electrically connected in parallel to the DC bus **516**, such as to a DC return bus of the DC bus **516**. The collector end of the first transistor **520***a* and the cathode end of the first diode **522***a* may be electrically connected in parallel to an inductor **514**, such as a ripple eliminating inductor, as a first end of the inductor **514**. The inductor **514**, at a second end, may be electrically connected to anode end of the capacitor **512** via the second DC bus **518**. The second DC bus **518** may be used to remove a ripple from the DC/DC converter **508**, and may not carry any active power. The

capacitor **512**, at a cathode end, may be electrically connected to the DC bus **516**, such as to a DC return bus of the DC bus **516**. The emitter end of the second transistor **520***b* and the anode end of a second diode **522***b* may be electrically connected in parallel to the first end of the inductor **512**. The collector end of a second transistor **520***b* and the cathode end of a second diode **522***b* may be electrically connected in parallel to DC bus **516**. The AC signals generated by the AC sweep generator **510** may be used by the DC/DC converter **508** to generate an impedance that may be compared against the impedance measured by the bias of the DC/DC converter **506** to generate a bias signal. The DC/DC converter **508** may be a bidirectional converter which is the comparison to the DC/DC converter **506** with which a basis for the measured impedance is formed. The capacitor **512** removes the ripple from the DC/DC converter **508**.

(50) In some embodiments, the power electronics hardware **100***a*, **100***b*, **200**, **300***a*, **300***b*, **400**, and **500** may be unidirectional. In some embodiments, the power electronics hardware **100***a*, **100***b*, **200**, **300***a*, **300***b*, **400**, and **500** may be bidirectional.

(51) In one embodiment, an alternating current (AC) impedance spectroscopy method comprises providing an AC impedance spectroscopy ripple from power electronics into an electrochemical device, and absorbing the ripple in the power electronics.

(52) In one embodiment, the AC impedance spectroscopy ripple is provided from power electronics into the electrochemical device while an operating current or voltage is provided to or from the electrochemical device, such that the AC impedance spectroscopy ripple does interrupt the operation of the electrochemical device.

(53) In one embodiment, the electrochemical device comprises an electrochemical stack, and impedance of different portions of the electrochemical stack is measured separately and compared to each other or to a reference or average impedance value. The method may also include determining a fault in a portion of the electrochemical stack, and electrically bypassing the portion of electrochemical stack containing the fault. The electrochemical stack may comprise a proton exchange membrane (PEM) electrolyzer stack.

(54) In one embodiment, the power electronics comprise a plurality of electric power inverters electrically connectable to respective AC busses of the AC split bus, a first capacitor and a second capacitor electrically connected in parallel to the plurality of electric power inverters via a middle DC bus at a DC-link midpoint, and an auxiliary electric power converter electrically connected to the DC-link midpoint via the middle DC bus and electrically connectable to the direct current (DC)/DC converter via the DC bus. In one embodiment, the auxiliary electric power converter is configured to draw a reactive power amount from the DC-link to cancel the AC impedance spectroscopy ripple. In one embodiment, the method also includes measuring the AC impedance spectroscopy ripple voltage of the DC-link based on voltages at the first capacitor and the second capacitor, and injecting a canceling ripple to the DC-link configured to remove the ripple voltage from the DC-link.

(55) With regard to FIG. **6**, the method **600** may also include outputting an AC voltage from the power electronics at varying frequencies, measuring an impedance at varying frequencies of the electrochemical device which comprises a proton exchange membrane (PEM) stack, determining whether the measured impedance exceeds a threshold for the AC voltage at a first frequency of the varying frequencies, determining a fault associated with the AC voltage at the first frequency in response to determining that the measured impedance exceeds the threshold for the AC voltage at the first frequency, and remedying the fault.

(56) FIG. **6** is a process flow diagram illustrating an example method **600** of impedance monitoring of a modular electrolysis system according to an embodiment. With reference to FIGS. **1**A-**5**B, the method **600** may be implemented in dedicated hardware, software configured to cause a processor (e.g., a controller) to execute instructions of the software, and/or a combination of dedicated hardware and software executing on a processor. The software may be stored on a non-transitory processor readable medium, such as a volatile memory (e.g., cache, random access memory, buffer,

etc.) or nonvolatile memory (fixed or removable storage memory, disk memory, solid state memory, etc.) and loaded to the processor for execution. For example, the method **600** may be implemented as a processor (e.g., controller) executing software within an AC impedance spectroscopy system (e.g., power electronics hardware **100***a*, **100***b*, **200**, **300***a*, **300***b*, **400**, and **500** in FIGS. **1**A-**2**A, **3**A-**4**A, **5**A, and **5**B, power electronics hardware control **210**, **410** in FIGS. **2**B and **4**B) that includes other individual components, and various memory/cache controllers. In order to encompass the alternative configurations enabled in various embodiments, the hardware implementing the method **600** is referred to herein as an “processing device.”
(57) In block **602**, the processing device may generate and output AC voltage at varying frequencies. The processing device may control various hardware components (e.g., transistors **116***a*, **116***b* in FIGS. **1**A-**2**A, transistors **306***a*, **306***b*, **308***a*, **308***b* in FIGS. **3**A-**4**A, and/or AC sweep generator **510** in FIG. **5**A) generate and output AC voltage at varying frequencies. The processing device sweeps through a set of preferred frequencies or an entire spectrum of frequencies. In some embodiments, the AC voltage frequencies may be between approximately 1 Hz and approximately 10 kHz. A maximum frequency of a perturbation may be within a bandwidth of a power electronic converter (e.g., DC/DC converter **104** in FIGS. **1**A-**2**A and **3**A-**4**A, auxiliary electric power converter **110** in FIGS. **1**A, **1**B, **3**A, and **3**B, and/or DC/DC converter **506** in FIG. **5**A).
(58) In block **604**, the processing device may measure impedance resulting from output of the AC voltage at varying frequencies. The processing device may take measurements of the resultant impedance of the AC voltages going through specified systems, such as a cell as a power source and/or a cell as a load. Measurements may be made by the processing device by monitoring voltages and currents at different locations of an AC impedance spectroscopy system (e.g., power electronics hardware **100***a*, **100***b*, **200**, **300***a*, **300***b*, **400**, and **500** in FIGS. **1**A-**2**A, **3**A-**4**A, **5**A, and **5**B, power electronics hardware control **210**, **410** in FIGS. **2**B and **4**B), such as at a power electronic converter (e.g., DC/DC converter **104** in FIGS. **1**A-**2**A and **3**A-**4**A, auxiliary electric power converter **110** in FIGS. **1**A, **1**B, **3**A, and **3**B, DC/DC converter **506** in FIG. **5**A).
(59) In determination block **606**, the processing device may determine whether a measurement of impedance exceeds a threshold for the AC voltage frequency. The processing device may access and retrieve threshold values of impedance for AC voltage frequencies from a memory, stored in association with each other, such as in a data structure or a database. The processing device may compare a measurement of impedance for the AC voltage frequency with the threshold values of impedance for AC voltage frequency to determine whether the measurement of impedance exceeds a threshold for the AC voltage frequency.
(60) In response to determining that the measurement of impedance does not exceed a threshold for the AC voltage frequency (i.e., determination block **606**=“No”), the processing device may generate and output AC voltage at varying frequencies in block **602**. In response to determining that the measurement of impedance does exceed a threshold for the AC voltage frequency (i.e., determination block **606**=“Yes”), the processing device may determine and handle a fault associated with the AC voltage frequency in block **608**. The processing device may access and retrieve fault information, including fault type, processes for remedying or addressing the fault, etc. for the AC voltage frequency from a memory, stored in association with each other, such as in a data structure or a database. The processing device may implement the processes for remedying or addressing the fault.
(61) The following are non-exhaustive examples of faults and processes for remedying the faults for different voltage frequencies. For example, a fault associated with a first voltage frequency may include dryout detection of the PEM, and the fault may be handled by the processing device causing an increase in a humidification function in order to increase inlet stream water content to the PEM stack in order to properly humidify and moisturize the PEM. In some embodiments, each time the fault occurs, the processing device may take account of an environment in and/or around the PEM stack and may run humidification processes increasingly more often and/or at higher

levels. In one embodiment, flow direction of a dead-ended PEM stack may be reversed (inlet to outlet reverses to outlet to inlet) by the processing device when PEM dryout is detected via impedance in the range of approximately 1 kHz.
(62) In another example, a fault associated with a second voltage frequency may include poisoning of the PEM. If the fault occurs where there are impurities in the input constituents (e.g., water, pure oxygen) and the PEM is poisoned, a change in the cell impedance spectra may be detected. In some embodiments, the processing device may handle this fault by triggering cell rejuvenation cycles, such as hydrogen pumping from one electrode to another, or an oxidation/reduction cycle of the electrodes. In some embodiments, the processing device may handle this fault by triggering an increased function of impurity filtering mechanisms in the electrochemical system such as water or air or fuel purification.
(63) In another example, a fault associated with a third voltage frequency may include flooding (e.g., accumulation of liquid water) of the PEM, and the fault may be handled by the processing device causing the PEM system controls to execute either a purge of the chambers where flooding is occurring, such as on the anode or cathode of the cell, and/or increasing a rate of recirculation of the anode or cathode to clear the flooding condition. In one embodiment, flooding fault may be detected by the processing device by monitoring the impedance of the PEM stack at a frequency of 1 Hz.
(64) In another example, a fault associated with a fourth voltage frequency may include leakage from the PEM. If a leak occurs, such as detected by the processing device when there is a divergence between electrochemical system function (output flows or output power per input flow) as compared with the baseline impedance spectra, the leakage fault may be handled by the processing device by causing the electrochemical system to adjust and supply more fuel (not running as efficiently) to cover needs. The processing device may send an error code for service in order for the leak to be fixed. An impedance measurement may be taken to determine where the exact leak occurred.
(65) In another example, a fault associated with a fifth voltage frequency may include excess reverse diffusion. If excess reverse diffusion occurs, such as detected by the processing device when there is a divergence of output as compared to cell impedance values, then the electrochemical system will evaluate why the excess happens, and if it is strictly environmental, the excess reverse diffusion fault may be handled by the processing device by causing the electrochemical system to correct itself and decrease input reactant use to minimize reverse diffusion.
(66) In another example, a fault associated with a sixth voltage frequency may include bubble formation on an anode. If bubbles form on the anode plate as detected by the processing device through low frequency impedance monitoring, the processing device may handle this fault by jogging or increasing water flow in order to remove the bubbles.
(67) In another example, a fault associated with a seventh voltage frequency may include insufficient compression. The processing device may detect an insufficient compression fault when impedance spectra indicate that bulk resistance has increased. The processing device may handle this fault by increasing compression may in order to compensate and correct the condition.
(68) In block **610**, the processing device may send notice(s) to a service team. In some embodiments, the processing device may send a notice that a fault has occurred. The notice may be specific as to the type of fault detected and/or include AC impedance spectroscopy results, such as an impedance measurement, a voltage frequency, and/or a location in the electrochemical system of the measured impedance. In some embodiments, the processing device may send a notice that the error has been resolved and/or not resolved. In some embodiments, the processing device may send a notice of the impact on the electrochemical system performance.
(69) In some embodiments, the blocks of the method **600** may be implemented in any combination of continuously, periodically, and responsively. For example, blocks **602**, **604**, **606** may be

continuously implemented, even in parallel with the implementation of each other and/or other blocks of the method **600**.

(70) In some embodiments, the impedance spectroscopy is used to separately measure the impedance of portions of an electrochemical stack, such as a PEM stack. For example, impedance of one portion of a stack may be compared to an impedance of another portion of the same stack, or to a reference impedance or the average of the stack with impedance values. A differential impedance measurement across portion of the stack may pinpoint the error or fault easier. Furthermore, in addition to the measuring of impedance, the triggering current bypass or shunting of portion (i.e., section) sections of a stack based on impedance will also occur. This provides for a more efficient system which requires less manual service and maintains a longer lifespan. In other words, a defective portion of a PEM stack may be bypassed electrically in case of an error impedance measurement from the defective portion.

(71) In one embodiment, the power electronics hardware **100***a*, **100***b*, **200**, **300***a*, **300***b*, **400**, and **500** in FIGS. **1**A-**2**A, **3**A-**4**A, **5**A, and **5**B, power electronics hardware control **210**, **410** in FIGS. **2**B and **4**B place the AC impedance spectroscopy ripple into an electrochemical device, and simultaneously absorbs the ripple so that they can be placed into a DC system or other non-bipolar system without interrupting an operating current or voltage being provided to or from the electrochemical device. In one embodiment, a ripple of complex waveforms, such as multiple superimposed ripples or square waves, may be provided in order to create multiple frequencies for analysis with a single injected ripple waveform.

(72) Various examples (including, but not limited to, the examples discussed above with reference to FIGS. **1**A-**6**) may also be implemented in computing systems, such as any of a variety of commercially and/or proprietary integrated or embedded computing systems. An example computing system **700** is illustrated in FIG. **7**. Such a computing system **700** typically includes one or more processor assemblies **701** (e.g., a controller, a CPU, a DSP, an FPGA, an ASIC, etc.) coupled to volatile memory **702** and a large capacity nonvolatile memory **704** (e.g., a disk drive, solid state drive, etc.). The computing system **700** may also include a data port or drive **706** (e.g., a compact disc (CD) or digital versatile disc (DVD) disc drive, a USB port, a serial port, etc.) coupled to the processor assemblies **701**. The computing system **700** may also include network access ports **703** coupled to the processor assemblies **701** for establishing network interface connections with a network **705**, such as a local area network or wide area network coupled to other broadcast system computers and servers, the Internet, the public switched telephone network, and/or a cellular data network.

(73) With reference to FIGS. **1**A-**7**, the processor assemblies **701** may be any programmable microprocessor, microcomputer, or multiple processor chip or chips that can be configured by software instructions (applications) to perform a variety of functions, including the functions of various examples described above. In some devices, multiple processors may be provided, such as one processor dedicated to wireless communication functions and one processor dedicated to running other applications. Typically, software applications may be stored in the internal memory **702** before they are accessed and loaded into the processor assemblies **701**. The processor assemblies **701** may include internal memory sufficient to store the application software instructions. In many devices the internal memory **702** may be a volatile or nonvolatile memory, such as flash memory, or a mixture of both. For the purposes of this description, a general reference to memory refers to memory accessible by the processor assemblies **701**, including internal memory **702** or removable memory plugged into the device and memory **702** within the processor assemblies **701**, themselves.

(74) The above systems, devices, methods, processes, and the like may be realized in hardware, software, or any combination of these suitable for the control, data acquisition, and data processing described herein. This includes realization in one or more microprocessors, microcontrollers, embedded microcontrollers, programmable digital signal processors or other programmable devices

or processing circuitry, along with internal and/or external memory. This may also, or instead, include one or more application specific integrated circuits, programmable gate arrays, programmable array logic components, or any other device or devices that may be configured to process electronic signals. It will further be appreciated that a realization of the processes or devices described above may include computer-executable code created using a structured programming language such as C, an object oriented programming language such as C++, or any other high-level or low-level programming language (including assembly languages, hardware description languages, and database programming languages and technologies) that may be stored, compiled or interpreted to run on one of the above devices, as well as heterogeneous combinations of processors, processor architectures, or combinations of different hardware and software. At the same time, processing may be distributed across devices such as the various systems described above, or all of the functionality may be integrated into a dedicated, standalone device. All such permutations and combinations are intended to fall within the scope of the present disclosure.
(75) Embodiments disclosed herein may include computer program products comprising computer-executable code or computer-usable code that, when executing on one or more computing devices, performs any and/or all of the steps of the control systems described above. The code may be stored in a non-transitory fashion in a computer memory, which may be a memory from which the program executes (such as random access memory associated with a processor), or a storage device such as a disk drive, flash memory or any other optical, electromagnetic, magnetic, infrared or other device or combination of devices. In another aspect, any of the control systems described above may be embodied in any suitable transmission or propagation medium carrying computer-executable code and/or any inputs or outputs from same.
(76) The method steps of the implementations described herein are intended to include any suitable method of causing such method steps to be performed, consistent with the patentability of the following claims, unless a different meaning is expressly provided or otherwise clear from the context. So, for example performing the step of X includes any suitable method for causing another party such as a remote user, a remote processing resource (e.g., a server or cloud computer) or a machine to perform the step of X. Similarly, performing steps X, Y and Z may include any method of directing or controlling any combination of such other individuals or resources to perform steps X, Y and Z to obtain the benefit of such steps. Thus, method steps of the implementations described herein are intended to include any suitable method of causing one or more other parties or entities to perform the steps, consistent with the patentability of the following claims, unless a different meaning is expressly provided or otherwise clear from the context. Such parties or entities need not be under the direction or control of any other party or entity, and need not be located within a particular jurisdiction.
(77) It will be appreciated that the methods and systems described above are set forth by way of example and not of limitation. Numerous variations, additions, omissions, and other modifications will be apparent to one of ordinary skill in the art. In addition, the order or presentation of method steps in the description and drawings above is not intended to require this order of performing the recited steps unless a particular order is expressly required or otherwise clear from the context. Thus, while particular embodiments have been shown and described, it will be apparent to those skilled in the art that various changes and modifications in form and details may be made therein without departing from the scope of the disclosure.

Claims

1. An alternating current (AC) impedance spectroscopy system, comprising: a power factor corrected (PFC) rectifier electrically connectable to an AC power source via an AC split bus and to a direct current DC/DC converter via a first DC bus, wherein the PFC rectifier includes: a plurality of electric power inverters electrically connectable to respective AC busses of the AC split bus; a

first capacitor and a second capacitor electrically connected in parallel to the plurality of electric power inverters via a middle DC bus at a DC-link midpoint; and an auxiliary electric power converter electrically connected to the DC-link midpoint via the middle DC bus and electrically connectable to the direct current DC/DC converter via the first DC bus, the auxiliary electric power converter configured to draw a reactive power amount from the DC-link to cancel a ripple current drawn by the DC/DC converter and comprising: a third capacitor electrically connected in parallel to the first capacitor and a fourth capacitor electrically connected in parallel to the second capacitor; and a filtering inductor electrically connected to the third capacitor and the fourth capacitor in parallel, wherein current from the DC-link midpoint is alternately and controllably drawn through the filtering inductor or the third capacitor and the fourth capacitor.

2. The AC impedance spectroscopy system of claim 1, wherein the auxiliary electric power converter includes: the filtering inductor having a first end and a second end, wherein the filtering inductor is electrically connected at the first end to the third capacitor and the fourth capacitor in parallel; and a half-bridge inverter electrically connected to the second end of the filtering inductor.

3. The AC impedance spectroscopy system of claim 2, wherein the half-bridge inverter comprises: a first transistor electrically connected to the second end of the filtering inductor at an emitter end; a first diode electrically connected to the second end of the filtering inductor at an anode end and in parallel with the first transistor; a second transistor electrically connected to the second end of the filtering inductor at a collector end; and a second diode electrically connected to the second end of the filtering inductor at a cathode end and in parallel with the second transistor.

4. The AC impedance spectroscopy system of claim 2, wherein: the third capacitor is electrically connected to the first DC bus at a first end and to the middle DC bus and the first end of the filtering inductor at a second end; the fourth capacitor is electrically connected to the middle DC bus and the first end of the filtering inductor at a first end, and to a return DC bus at a second end; and the half-bridge inverter comprises: a first transistor electrically connected to the first DC bus at a collector end; a first diode electrically connected to the first DC bus at a cathode end and in parallel with the first transistor; a second transistor electrically connected to the return DC bus at an emitter end; and a second diode electrically connected to the return DC bus at an anode end and in parallel with the second transistor.

5. The AC impedance spectroscopy system of claim 1, wherein the plurality of electric power inverters each comprise: a first diode and a second diode electrically connected to the respective AC bus of the split AC bus in parallel, wherein the first diode is electrically connected to the respective AC bus at an anode end and the second diode is electrically connected to the respective AC bus at a cathode end; a first transistor comprising a collector end and an emitter end, the collector end electrically connected to the respective AC bus; a third diode comprising a cathode end and an anode end, the cathode end electrically connected to the respective AC bus at and the anode end electrically connected to the emitter end of the first transistor; a second transistor comprising a collector end and an emitter end, the collector end electrically connected to the middle DC bus and the emitter end electrically connected to the emitter end of the first transistor; and a fourth diode comprising a cathode end and an anode end, the cathode end electrically connected to the middle DC bus at a cathode end and the anode end electrically connected to the emitter end of the second transistor.

6. The AC impedance spectroscopy system of claim 1, comprising a controller configured with controller executable instructions to cause the controller to implement operations comprising: measuring a ripple voltage of the DC-link based on voltages at the first capacitor and the second capacitor in response to injection of a perturbation to the DC/DC converter; and injecting a canceling ripple to the DC-link configured to remove the ripple voltage from the DC-link.

7. The AC impedance spectroscopy system of claim 1, wherein the auxiliary electric power converter includes a third capacitor and a fourth capacitor electrically connected in parallel to the middle DC bus; and further comprising a controller configured with controller executable

instructions to cause the controller to implement operations comprising: measuring a ripple voltage of the DC-link based on voltages at the third capacitor and the fourth capacitor in response to injection of a perturbation to the DC/DC converter and in response to the first capacitor and the second capacitor being inaccessible; and injecting a canceling ripple to the DC-link configured to remove the ripple voltage from the DC-link.

8. The AC impedance spectroscopy system of claim 1 further comprising one or more additional power factor corrected (PFC) rectifiers electrically connectable to the AC power source via the AC split bus and to the direct current DC/DC converter via the first DC bus.

9. The AC impedance spectroscopy system of claim 8, wherein the one or more PFC rectifiers are connected in parallel to the PFC rectifier, the one or more PFC rectifiers configured to cancel one or more additional ripple currents drawn by the DC/DC converter.